

Research Profile

Physicist working at the interface of **reservoir computing, nonlinear dynamical systems, complex systems, and adaptive computation**. My research investigates how **hidden internal states, multiple timescales, contextual interactions, and collective dynamics** generate memory, adaptation, nonreciprocity, history-dependent response, and computational function. Recent work spans reservoir computing, non-equilibrium collective dynamics, scientific machine learning, neural and physiological time-series analysis, topological data analysis, and contextual inference.

Reservoir Computing

Nonlinear Dynamics

Complex Systems

Scientific ML

Collective Dynamics

EEG / Biosignals

Topological Data Analysis

Contextual Inference

Core Expertise

Reservoir Computing & Adaptive Computation

Reservoir architectures; recurrent dynamical systems; multi-timescale memory; controlled dissipation; machine learning and scientific ML

Numerical & Analytical Modelling

Numerical simulation; analytical modelling; scientific computing; dynamical-system analysis; high-dimensional time-series methods

Nonlinear & Complex Dynamical Systems

Nonlinear and multiscale dynamics; non-equilibrium systems; collective dynamics; memory, hysteresis, and nonreciprocal interactions

Neural Signals, Geometry & Topology

EEG and physiological signal analysis; persistent homology; graph methods; topological data analysis and discrete geometry

Education

Ph.D., High Energy Physics

Homi Bhabha National Institute

Harish-Chandra Research Institute, Prayagraj

Aug 2012–Sep 2018

Awarded Infosys Scholarship, 2017

Integrated M.Sc. Physics

Indian Institute of Technology Roorkee

July 2007–Aug 2012

Awarded Institute Silver Medal, 2012

Professional Experience

Chief Scientist

ManasAI Technology Pvt. Ltd.

Mar 2026–Present

EEG and Cognitive Neuroscience SME

Brainwave Science Pvt. Ltd.

Sep 2025–Feb 2026

Principal Investigator

Center for Philosophical and Cognitive Sciences, ISS Delhi

Research centre supported through the IKS Division, Ministry of Education, Government of India.

Dec 2022–Present

Institute Postdoctoral Fellow

Indian Institute of Technology Mandi, Himachal Pradesh

Jul 2022–Jul 2025

Career Break

Returned to full-time academic research in 2022 due to family/person issues.

Sep 2018 – July 2022

Publications

Reservoir Computing and Adaptive Computation

Preprints / under review

- Jyotiranjana Beuria** and Venkatesh H. Chembrolu. “Feasibility and Memory Mechanisms of Chern–Simons Context Reservoir Computation.” 2026. [arXiv:2609.13315](https://arxiv.org/abs/2609.13315). Under review.
- Jyotiranjana Beuria** and Amit Shukla. “Lindblad-Inspired Multi-Timescale Reservoir Computing with Separable Rotation and Dissipation.” 2026. [doi:10.48550/arXiv.2608.04028](https://doi.org/10.48550/arXiv.2608.04028). Under review.

Complex Systems and Collective Dynamics

Published

- Jyotiranjana Beuria** and Venkatesh H. Chembrolu. “Topological Flux on a Context Manifold Generates Nonreciprocal Collective Dynamics.” *Physical Review E* 113(6), 065401 (2026). [doi:10.1103/PhysRevE.113.065401](https://doi.org/10.1103/PhysRevE.113.065401).
- Jyotiranjana Beuria**, Mayank Chaurasiya, and Laxmidhar Behera. “Collective motion from quantum-inspired dynamics in visual perception.” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 481(2321), 20250489 (2025). [doi:10.1098/rspa.2025.0489](https://doi.org/10.1098/rspa.2025.0489).
- Jyotiranjana Beuria** and Laxmidhar Behera. “Non-local interaction in discrete Ricci curvature-induced biological aggregation.” *Royal Society Open Science* 11(9), 240794 (2024). [doi:10.1098/rsos.240794](https://doi.org/10.1098/rsos.240794).

Preprints / under review

6. **Jyotiranjana Beuria**. “Self-Healing Coordination in Cognitive Swarm Agents with Bloch-Type Perceptual Memory.” 2026. [doi:10.48550/arXiv.2607.11960](https://doi.org/10.48550/arXiv.2607.11960) . Under review.
7. **Jyotiranjana Beuria**. “Non-Markovian collective motion from self-regulated perceptual dynamics.” 2025. [doi:10.48550/arXiv.2510.23688](https://doi.org/10.48550/arXiv.2510.23688) . Under review.

Computational Cognition and AI

Published

8. **Jyotiranjana Beuria**. “Emergent Non-Classical Probabilistic Structure in Large Language Models Under Contextual Modulations.” *Scientific Reports* (2026). [doi:10.1038/s41598-026-65824-7](https://doi.org/10.1038/s41598-026-65824-7) .
9. Venkatesh H. Chembrolu, Vasudeva Prabhath Lolugu, and **Jyotiranjana Beuria**. “Beyond Alignment: Representational Ethics and the Governance of Constructed Worlds.” *AI and Ethics* 6, 508 (2026). [doi:10.1007/s43681-026-01355-1](https://doi.org/10.1007/s43681-026-01355-1) .

Preprints / under review

10. **Jyotiranjana Beuria**, Venkatesh H. Chembrolu, and Nitin Kumar. “A Formal Framework for Phenomenal Unity, Contextual Experience, and Intersubjective Coherence.” 2026. *PhilPapers*. Under review.
11. Lolugu Vasudeva Prabhath, Nitin Kumar, **Jyotiranjana Beuria**, and Venkatesh H. Chembrolu. “Semantic Information Beyond Shannon: Meaning, Coherence, and Agency in Experiential Systems.” Preprint. [doi:10.21203/rs.3.rs-10284367/v1](https://doi.org/10.21203/rs.3.rs-10284367/v1) .

Neural and Physiological Signal Analysis

Published

12. Swati Singh, **Jyotiranjana Beuria**, Laxmidhar Behera, Ram Bilas Pachori, and Vinay Gupta. “Decoding cognitive load changes induced by mantra meditation from physiological signals using deep neural networks.” *IEEE Sensors Journal* 26(1), 1088–1102 (2026). [doi:10.1109/JSEN.2025.3628683](https://doi.org/10.1109/JSEN.2025.3628683) .
13. Kurusetti Vinay Gupta, **Jyotiranjana Beuria**, Lokeshwara Kumar Vijanapalli, Amit Sethi, and Laxmidhar Behera. “Self-reflection, sense of agency, and underlying neural correlates: A pilot study.” *PLOS ONE* 20(12), e0335276 (2025). [doi:10.1371/journal.pone.0335276](https://doi.org/10.1371/journal.pone.0335276) .
14. Kurusetti Vinay Gupta, **Jyotiranjana Beuria**, and Laxmidhar Behera. “Characterizing EEG signals of meditative states using persistent homology and Hodge spectral entropy.” *Biomedical Signal Processing and Control* 89, 105779 (2024). [doi:10.1016/j.bspc.2023.105779](https://doi.org/10.1016/j.bspc.2023.105779) .

Preprints / under review

15. Ashish Gupta, **Jyotiranjana Beuria**, and Braj Bhushan. “Guilt, Shame, and Remorse: Exploring EEG-based persistent homology of complex emotions.” 2026. Manuscript under review.
16. **Jyotiranjana Beuria**. “Topological and graph-theoretic analysis of EEG dynamics in mind-wandering and focused attention.” 2026. Manuscript under review.
17. Swati Singh, **Jyotiranjana Beuria**, Ram Bilas Pachori, and Laxmidhar Behera. “Neural and affective markers of kirtan meditation: Persistent homology of multivariate EEG in real and immersive environments.” 2026. Manuscript under review.
18. Durgesh Ameta, Lovepreet Singh, **Jyotiranjana Beuria**, and Laxmidhar Behera. “Multimodal Odor Perception Prediction Using Olfactory EEG and Physicochemical Features of Odorants.” *ChemRxiv* (2025). [doi:10.26434/chemrxiv-2025-fbmlf/v2](https://doi.org/10.26434/chemrxiv-2025-fbmlf/v2) . Preprint; manuscript under review.

High-Energy Physics and Collider Data Analysis (mostly during PhD)

Published

19. **Jyotiranjana Beuria**. “Intrinsic geometry of collider observations and Forman Ricci curvature.” *Physical Review D* 110(3), 035023 (2024). [doi:10.1103/PhysRevD.110.035023](https://doi.org/10.1103/PhysRevD.110.035023) .
20. **Jyotiranjana Beuria**. “Persistent homology of collider observations: When (w) hole matters.” *Physics Letters B* 846, 138188 (2023). [doi:10.1016/j.physletb.2023.138188](https://doi.org/10.1016/j.physletb.2023.138188) .
21. **Jyotiranjana Beuria**, AseshKrishna Datta, Dipsikha Debnath, and Konstantin T. Matchev. “LHC collider phenomenology of minimal universal extra dimensions.” *Computer Physics Communications* 226, 187–205 (2018). [doi:10.1016/j.cpc.2017.11.021](https://doi.org/10.1016/j.cpc.2017.11.021) .
22. **Jyotiranjana Beuria** and AseshKrishna Datta. “Spontaneous breakdown of charge in the MSSM and in the NMSSM: Possibilities and implications.” *Journal of High Energy Physics* 2017(11), 1–26 (2017). [doi:10.1007/JHEP11\(2017\)042](https://doi.org/10.1007/JHEP11(2017)042) .
23. **Jyotiranjana Beuria** and Abhishek Dey. “Exploring charge and color breaking vacuum in non-holomorphic MSSM.” *Journal of High Energy Physics* 2017(10), 1–27 (2017). [doi:10.1007/JHEP10\(2017\)154](https://doi.org/10.1007/JHEP10(2017)154) .
24. **Jyotiranjana Beuria**, Utpal Chattopadhyay, AseshKrishna Datta, and Abhishek Dey. “Exploring viable vacua of the Z_3 -symmetric NMSSM.” *Journal of High Energy Physics* 2017(4), 1–52 (2017). [doi:10.1007/JHEP04\(2017\)024](https://doi.org/10.1007/JHEP04(2017)024) .
25. **Jyotiranjana Beuria**, Arindam Chatterjee, and AseshKrishna Datta. “Sbottoms of natural NMSSM at the LHC.” *Journal of High Energy Physics* 2016(8), 1–31 (2016). [doi:10.1007/JHEP08\(2016\)004](https://doi.org/10.1007/JHEP08(2016)004) .
26. **Jyotiranjana Beuria**, Arindam Chatterjee, AseshKrishna Datta, and Santosh Kumar Rai. “Two light stops in the NMSSM and the LHC.” *Journal of High Energy Physics* 2015(9), 1–47 (2015). [doi:10.1007/JHEP09\(2015\)073](https://doi.org/10.1007/JHEP09(2015)073) .

Preprints / under review

27. **Jyotiranjana Beuria**. “Topological Landscapes of the BSM Higgs Sector.” 2025. [doi:10.48550/arXiv.2510.10900](https://doi.org/10.48550/arXiv.2510.10900) . Under review.
28. **Jyotiranjana Beuria**. “Probing Invisible Decay of Z' at Muon Collider with Topological Data Analysis and Machine Learning.” 2025. [doi:10.48550/arXiv.2509.19756](https://doi.org/10.48550/arXiv.2509.19756) . Under review.

Peer-Reviewed Book Chapters

1. Manish Sharma, Pushkar Chitranshi Srivastava, Parveen Kumar, **Jyotiranjan Beuria**, and Laxmidhar Behera. “Non-linear EEG Marker of Mind Wandering During Mantra Meditation Among Novice Meditators.” In *Cognitive Biomarkers and BCI Applications*, MBCC 2025, Studies in Rhythm Engineering, Springer, Cham, 2026, pp. 105–116. doi:10.1007/978-3-032-20634-3_10.
2. Parveen Kumar, Perna Singh, Lokeswara Kumar Vijanapalli, **Jyotiranjan Beuria**, and Lalan Kumar. “A Case Study on Yogic Perception: Insights From EEG Data Analysis.” In *Cognitive Biomarkers and BCI Applications*, MBCC 2025, Studies in Rhythm Engineering, Springer, Cham, 2026, pp. 134–144. doi:10.1007/978-3-032-20634-3_12.
3. Kurusetti Vinay Gupta, **Jyotiranjan Beuria**, and Laxmidhar Behera. “Neuroscience of Meditation for Holistic Well-Being.” In *Biological Measures of Well-Being*, edited by Colin R. Martin, Vinood B. Patel, Rajkumar Rajendram, and Victor R. Preedy, Academic Press, 2026, pp. 175–188. doi:10.1016/B978-0-443-28842-5.00001-6.

Software Expertise

- **Scientific Computing & Modelling:** C/C++, Python; numerical simulation of nonlinear dynamical systems, reservoir computing, machine learning, signal processing, and scientific data analysis.
- **Experimental Software:** Experience with stimulus-presentation and behavioral/EEG experiment pipelines, including response collection and event synchronization.
- **Web & App Development:** Full-stack development with PHP, Node.js, JavaScript/TypeScript; cross-platform applications using Electron, Flutter, and Tauri.

Research Grants and Leadership

IKS Research Center Grant, 2023–2025	INR 35.27 lakh (37,000 USD)
IKS Division, Ministry of Education, Government of India	
IKS Research Center Grant, 2025–2027	INR 10 lakh (9,100 USD)
IKS Division, Ministry of Education, Government of India	

Teaching Experience

Introduction to Topological Data Analysis	Neuroimaging Techniques and Signal Processing
Physics of Collective Motion	Introduction to Quantum Mechanics

Academic Service and Conference Organisation

- Organised the one-day conference “Unriddling Inference: From Pramāṇa Theory to Modern Logic and AI”, IIT Delhi, 12 April 2026.
- Organised “The Enigma of Perception: Perspectives from Indian Knowledge System”, IIT Delhi, 24 August 2025.
- Organising team, Mind, Brain and Consciousness Conference, IKSMHA Centre, IIT Mandi, 4–7 June 2025.
- Organising team, Mind, Brain and Consciousness Conference, IIT Mandi, 14–16 December 2023.
- Organised “Mind, Matter, and Consciousness: From Information to Meaning”, International Virtual Conference, 18–19 December 2021.

Selected Awards and Distinctions

- Winner, IKS Category, Himalayan Start-up Trek, IIT Mandi, 2026.
- Infosys Scholarship in High Energy Physics, Harish-Chandra Research Institute, 2017.
- GATE Physics 2012: All India Rank 13.
- JEST 2012: All India Rank 20.
- CSIR-NET Junior Research Fellowship 2011: All India Rank 38.
- Institute Silver Medal, Indian Institute of Technology Roorkee, 2012.
- WISE Scholarship, Deutscher Akademischer Austauschdienst (DAAD), Germany, 2010.

Referees

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